Diagrams to Draw

Every apparatus set-up and sketch graph from your three sets of planning notes, drawn the way a labelled exam diagram should look. Each one lists the points a marker looks for.

Rules for every diagram

- ✎ Label every piece of apparatus with a ruled leader line, and state capacities (250 cm^3 conical flask, 50.00 cm^3 burette, 100 cm^3 gas syringe).
- ✎ Show liquid levels with a horizontal line; draw solids as dots or shading.
- ✎ Seal what must be sealed (gas collection) and leave open what must be open (reflux).
- ✎ Show a safe heat source: heating mantle or water bath for organic liquids; Bunsen burner only for inorganic heating.
- ✎ Name what is inside each vessel, with its volume or mass where you know it.
- ✎ Draw a set-up that would actually work: no gaps for gas to leak, tubes ending under water, thermometers in the liquid, a vertical burette.
- ✎ For gas collection, show how the reactants are kept apart and then mixed, and how the gas is collected.
- ✎ Sketch graphs: both axes labelled with quantity / unit, best-fit lines, and the key point marked (V_{eq}, ΔT, t = 3.0 min).

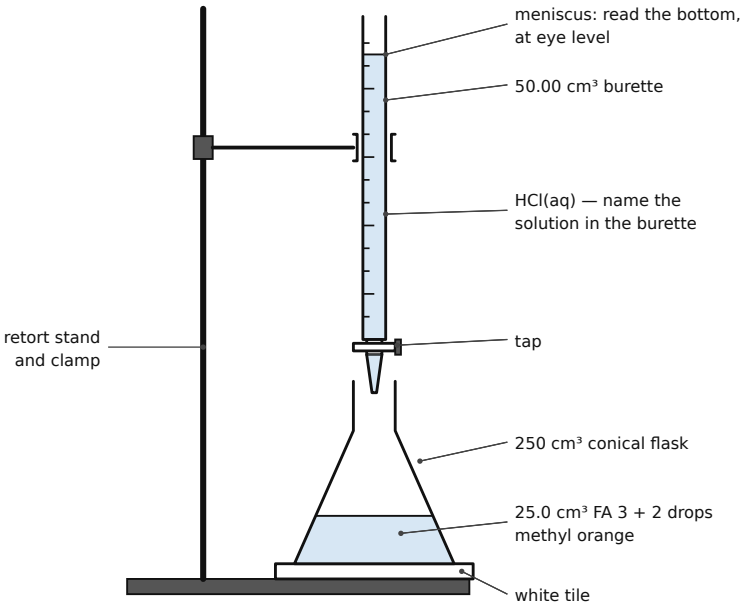
Contents

01	Titration set-up	Volumetric analysis
02	Making up a standard solution	Volumetric analysis
03	Micro-titration	Volumetric analysis
04	Heating a solid in a crucible	Gravimetry
05	Heating in a boiling tube	Gravimetry
06	Desiccator	Gravimetry
07	Gravity filtration	Gravimetry
08	Gas syringe set-up	Gas collection
09	Collecting over water: inverted measuring cylinder	Gas collection
10	Collecting over water: inverted burette	Gas collection
11	Calorimeter: polystyrene cup	Energetics
12	Thermometric titration	Energetics
13	Enthalpy change of combustion	Energetics
14	Iodine clock reaction	Kinetics
15	Separating funnel	Equilibria
16	Saturated solution at constant temperature	Equilibria
17	Electrolysis of molten $PbBr_2$	Electrochemistry
18	Electrochemical cell (Zn / Cu)	Electrochemistry
19	Heating under reflux	Organic synthesis
20	Distillation	Organic synthesis
21	Vacuum filtration	Organic synthesis
22	Cooling while adding a reagent dropwise	Organic synthesis
23	Mass against number of heatings	graph
24	Mass of precipitate against volume added	graph
25	Gas volume against time	graph
26	Temperature against time (exothermic)	graph
27	Temperature against time (endothermic)	graph
28	Thermometric titration	graph
29	Mixing varying volumes (total constant)	graph
30	1/t against V: finding the order	graph
31	V($KMnO_4$) against time	graph
32	Solubility against temperature	graph
33	E(cell) against ln(1/[Cu^{2+}])	graph

PART A

Apparatus set-ups

VOLUMETRIC ANALYSIS



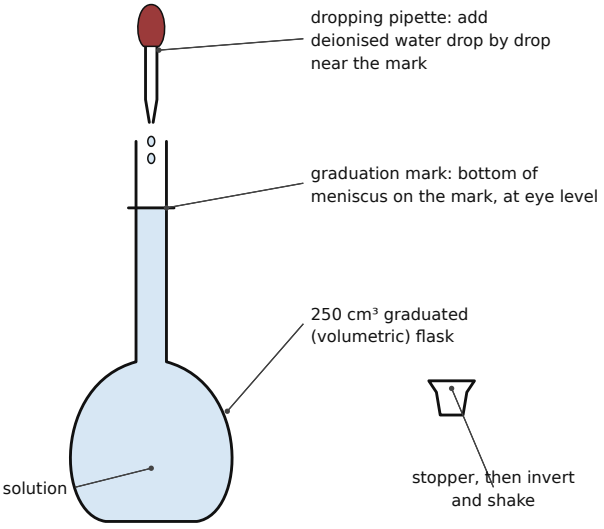
01 Titration set-up

Used in: Volumetric WE1, WE2 · Equilibria WE1, WE3 · Kinetics WE11

MUST SHOW

- Burette clamped **vertically**; tip just inside the neck of the flask.
- Label the solution in the burette **and** the solution (with volume) in the conical flask.
- Conical flask on a **white tile** (makes the colour change easy to see).
- Capacities: 50.00 cm³ burette, 250 cm³ conical flask.

Practise: cover the drawing and redraw it from the checklist.



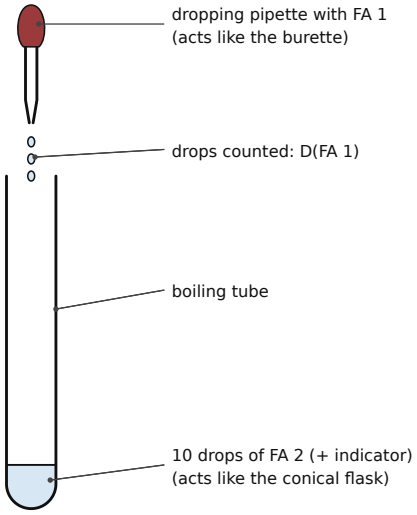
02 Making up a standard solution

Used in: Volumetric WE1 (solid) · WE2 and micro-titration (dilution)

MUST SHOW

- Neck with a single **graduation mark**; capacity stated (250 cm³ or 100 cm³).
- Liquid level at the mark (bottom of meniscus).
- Dissolve the solid in a beaker first; transfer with a funnel and glass rod, rinsing the washings in.
- For a dilution, run the stock solution in from a **burette** (or pipette).

Practise: cover the drawing and redraw it from the checklist.



03 Micro-titration

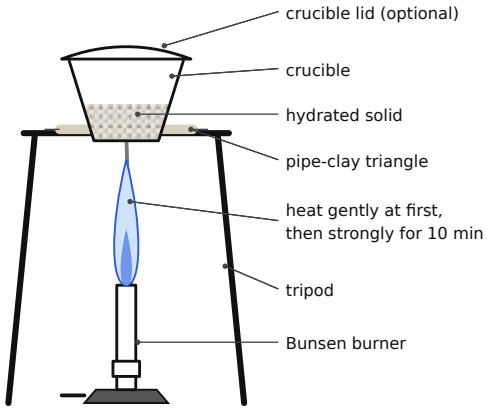
Used in: Volumetric: checking concentrations

MUST SHOW

- 10 drops of FA 2 in a boiling tube; FA 1 added from a dropping pipette.
- Rinse the dropper with deionised water, then with FA 1, before adding FA 1.
- Shake after each drop; count drops to the colour change.
- V(FA 1) ≈ D(FA 1)/10 × 25.0 cm³.

Practise: cover the drawing and redraw it from the checklist.

GRAVIMETRY



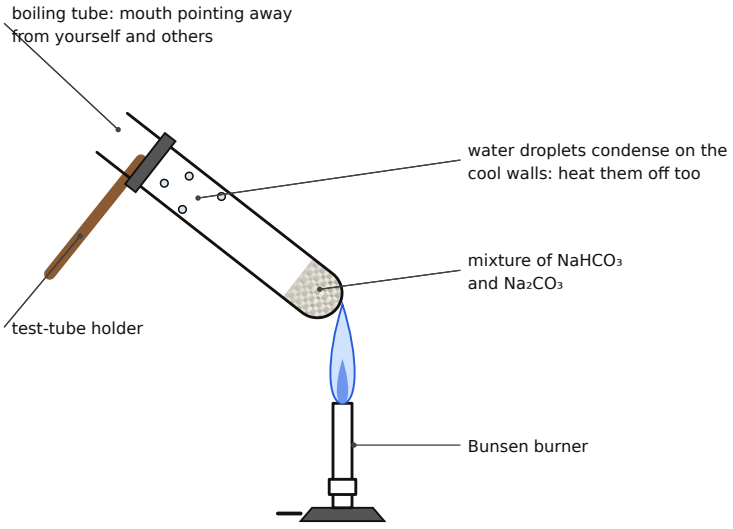
04 Heating a solid in a crucible

Used in: Gravimetry WE1 ($CuSO_4 \cdot nH_2O$) · Equilibria WE2 uses an evaporating dish the same way

MUST SHOW

- Crucible sitting **in a pipe-clay triangle** on a tripod.
- Bunsen burner directly underneath.
- Heat gently first (stops spattering), then strongly; then cool and weigh; repeat to constant mass.
- Use crucible tongs to handle the hot crucible.

Practise: cover the drawing and redraw it from the checklist.



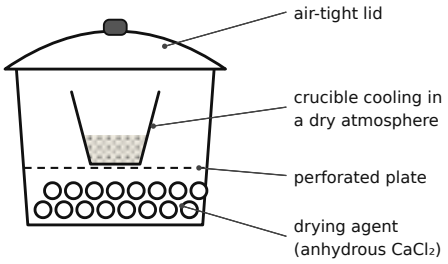
05 Heating in a boiling tube

Used in: Gravimetry WE2 ($NaHCO_3$ / Na_2CO_3)

MUST SHOW

- Tube held with a **test-tube holder**, tilted, mouth pointing away from people.
- Move the tube in the flame so it is not heated at one spot (prevents spurting).
- Condensed water on the cool upper walls must also be driven off.
- A crucible without a lid lets water vapour escape more easily.

Practise: cover the drawing and redraw it from the checklist.



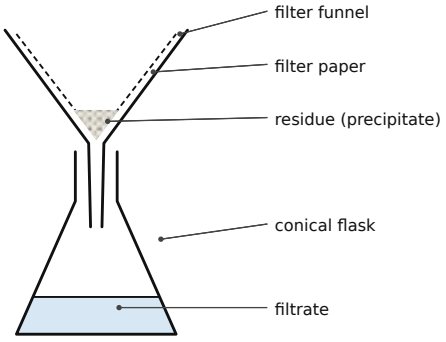
06 Desiccator

Used in: Gravimetry WE1, WE3 · Electrochemistry WE4

MUST SHOW

- Air-tight lid, perforated plate, drying agent underneath.
- Why: lets the hot crucible cool to room temperature without absorbing moisture.
- Weigh only when cool: hot air rising makes the reading too low.

Practise: cover the drawing and redraw it from the checklist.



07 Gravity filtration

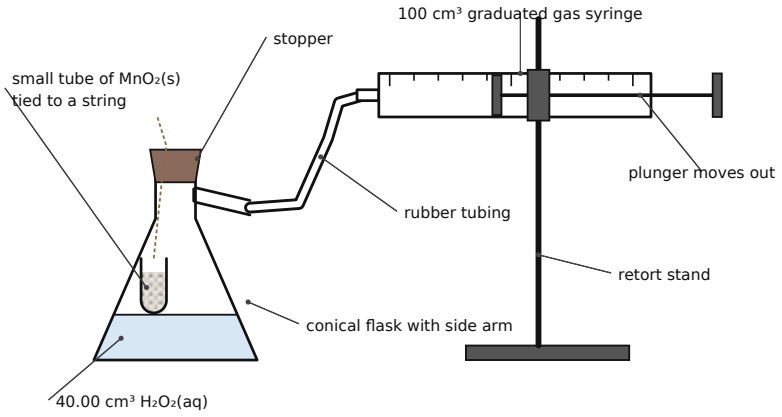
Used in: Gravimetry WE3 ($PbCl_2$) · Equilibria WE2, WE3 (dry apparatus)

MUST SHOW

- Funnel lined with folded filter paper; stem inside the flask neck.
- Label **residue** on the paper and **filtrate** below.
- Wash the precipitate with cold deionised water ($PbCl_2$ dissolves in hot water).
- For saturated solutions: dry funnel, dry filter paper, dry flask.

Practise: cover the drawing and redraw it from the checklist.

GAS COLLECTION



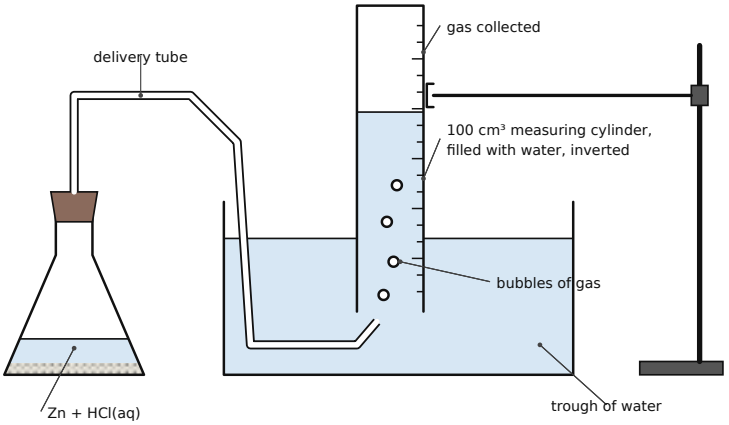
08 Gas syringe set-up

Used in: Gas collection WE4 (O_2 from H_2O_2), WE5 (CO_2 from Na_2CO_3)

MUST SHOW

- Solid in a **small tube tied to a string**, hung above the liquid so they do not mix until the stopper is loosened.
- Flask stoppered; side arm connected by rubber tubing to a **graduated gas syringe**, clamped.
- Syringe reads zero at the start; read again when the reading is constant.
- Capacity stated (100 cm^3); plan for 50–90% of it.

Practise: cover the drawing and redraw it from the checklist.



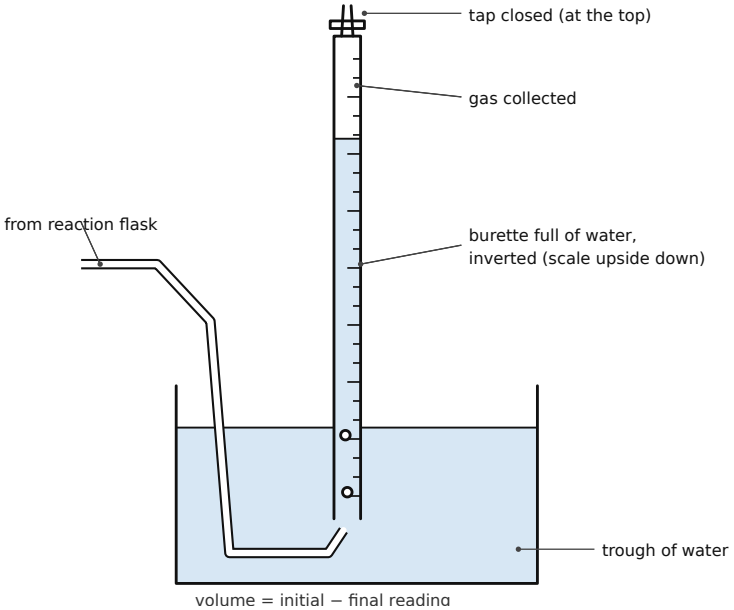
09 Collecting over water: inverted measuring cylinder

Used in: Gas collection: methods

MUST SHOW

- Cylinder full of water, inverted, mouth **under** the water in the trough; clamped.
- Delivery tube end under the mouth of the cylinder.
- Volume = final − initial reading.
- Only for gases that are insoluble or poorly soluble in water (not CO_2, NH_3, SO_2, NO_2, HX).

Practise: cover the drawing and redraw it from the checklist.



10 Collecting over water: inverted burette

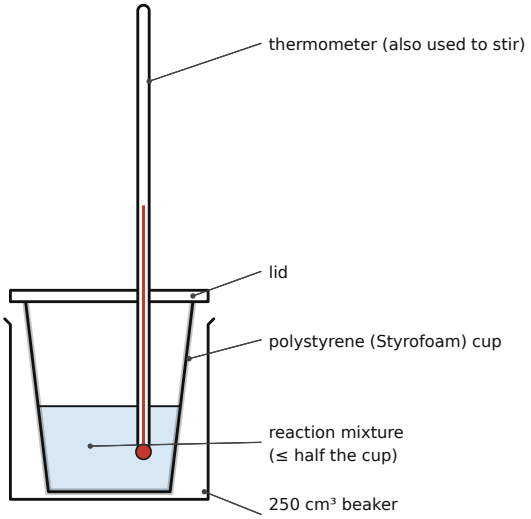
Used in: Gas collection: methods

MUST SHOW

- Burette filled with water, inverted, tap at the top closed.
- Readings fall as gas collects: volume = initial − final.
- Can only hold 50.00 cm³ of gas.

Practise: cover the drawing and redraw it from the checklist.

ENERGETICS



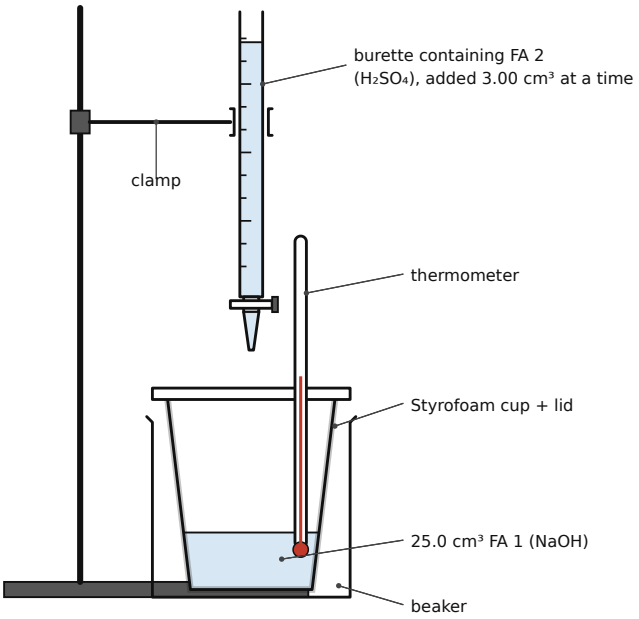
11 Calorimeter: polystyrene cup

Used in: Energetics WE6 (ΔH soln), temperature–time method, WE8

MUST SHOW

- Polystyrene cup **inside a 250 cm³ beaker**, with a lid.
- Thermometer through the lid, bulb in the solution.
- Solution no more than half the cup, so a solid can be added without spilling.

Practise: cover the drawing and redraw it from the checklist.



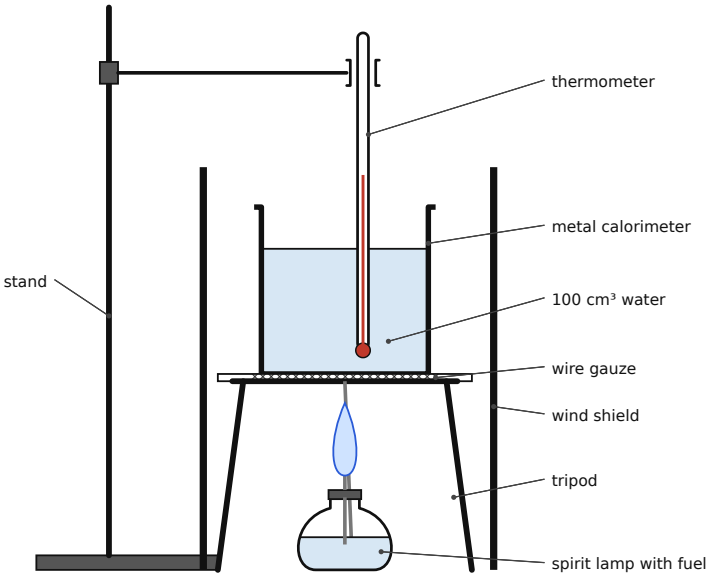
12 Thermometric titration

Used in: Energetics WE7

MUST SHOW

- Burette clamped above the cup; thermometer in the solution.
- Record the highest temperature after each 3.00 cm³ portion.
- Continue well past the end-point (≥ 5 readings after it).

Practise: cover the drawing and redraw it from the checklist.



13 Enthalpy change of combustion

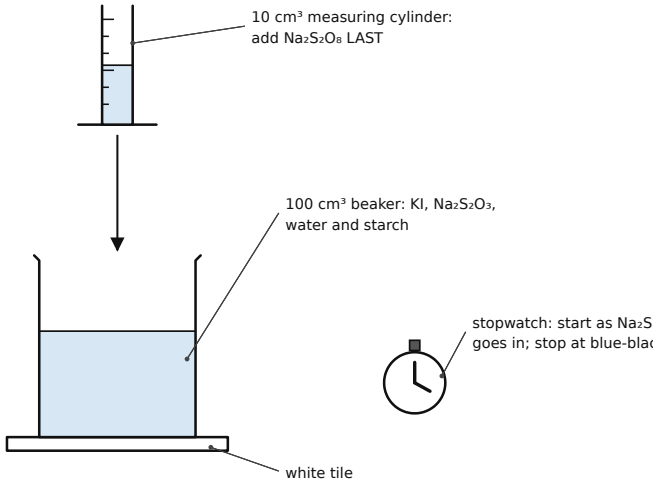
Used in: Energetics WE9 (5-decyne / cis-dec-5-ene; calibrate with hexane)

MUST SHOW

- Spirit lamp (wick partly in fuel) **under** the calorimeter.
- Calorimeter on a tripod and wire gauze; thermometer clamped and immersed in the water.
- Wind shield around the set-up.
- Weigh lamp + fuel before and after; stop when ΔT ≈ 5–10 °C.

Practise: cover the drawing and redraw it from the checklist.

KINETICS



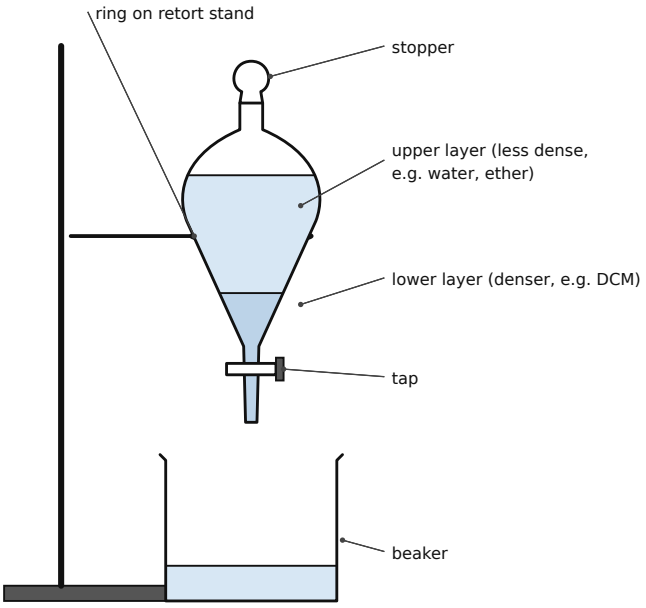
14 Iodine clock reaction

Used in: Kinetics WE10

MUST SHOW

- Total volume constant in every run (vary water).
- Last reagent added starts the stopwatch.
- White tile under the beaker to see the blue-black colour.

Practise: cover the drawing and redraw it from the checklist.



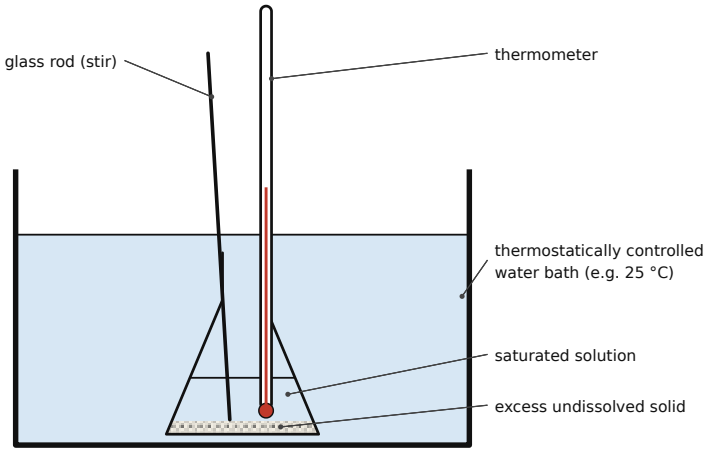
15 Separating funnel

Used in: Equilibria WE1 (K_D) · Organic WE6 (extraction)

MUST SHOW

- Pear-shaped funnel with stopper and tap, supported in a ring.
- Two layers: the **denser** one at the bottom (DCM 1.33 > water 1.00 > ether 0.71).
- Shake with the tap pointing up; open the tap briefly to release pressure.
- Remove the stopper before running off the lower layer.

Practise: cover the drawing and redraw it from the checklist.



16 Saturated solution at constant temperature

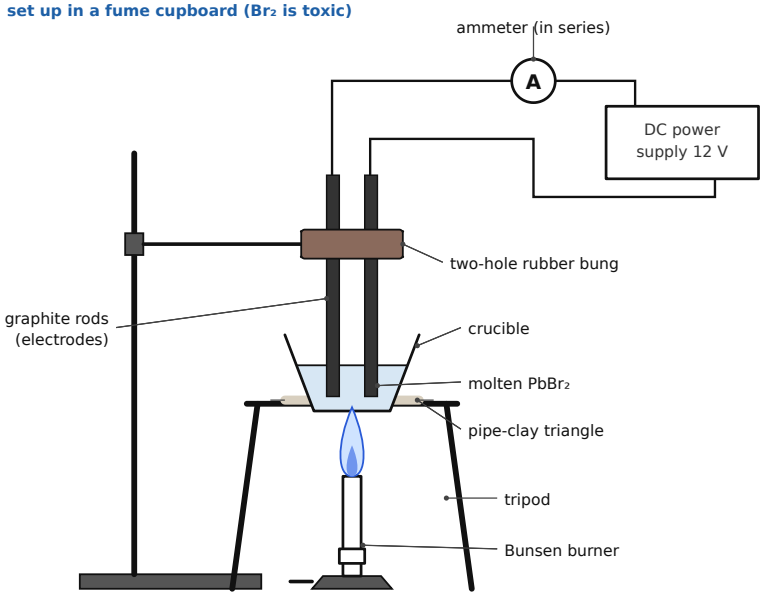
Used in: Equilibria WE2 ($Ce(SO_4)_2$), WE3 (KHT)

MUST SHOW

- Flask in a **thermostatically controlled** water bath, with a thermometer in the solution.
- Some solid left undissolved (shows the solution is saturated).
- Allow time (10–30 min) to reach equilibrium, then filter with dry apparatus.

Practise: cover the drawing and redraw it from the checklist.

set up in a fume cupboard (Br_2 is toxic)



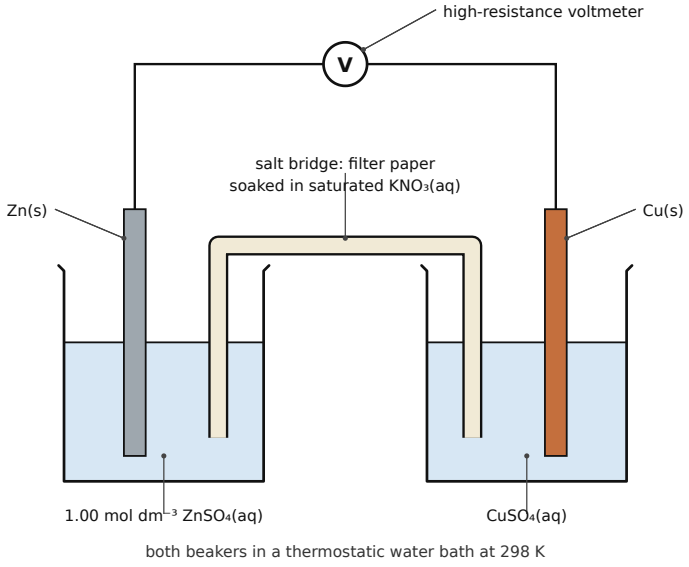
17 Electrolysis of molten $PbBr_2$

Used in: Electrochemistry WE4 (Avogadro's constant)

MUST SHOW

- Graphite rods through a two-hole bung, dipping into the $PbBr_2$ in a crucible.
- Ammeter in series with the DC supply.
- Crucible in a pipe-clay triangle on a tripod, heated by a Bunsen ($PbBr_2$ melts at 373 °C).
- Fume cupboard; handle the hot crucible with tongs.

Practise: cover the drawing and redraw it from the checklist.



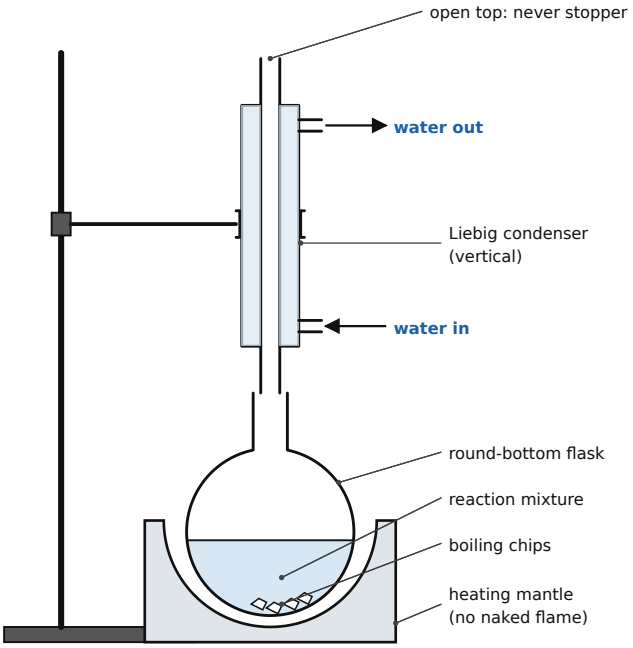
18 Electrochemical cell (Zn / Cu)

Used in: Electrochemistry WE5 (Nernst)

MUST SHOW

- Two half-cells: metal electrode in a solution of its own ions.
- **Salt bridge** (filter paper soaked in saturated KNO_3) dipping into both solutions.
- **High-resistance voltmeter** connected across the electrodes.
- Same temperature for both (298 K); new salt bridge for each reading.

Practise: cover the drawing and redraw it from the checklist.



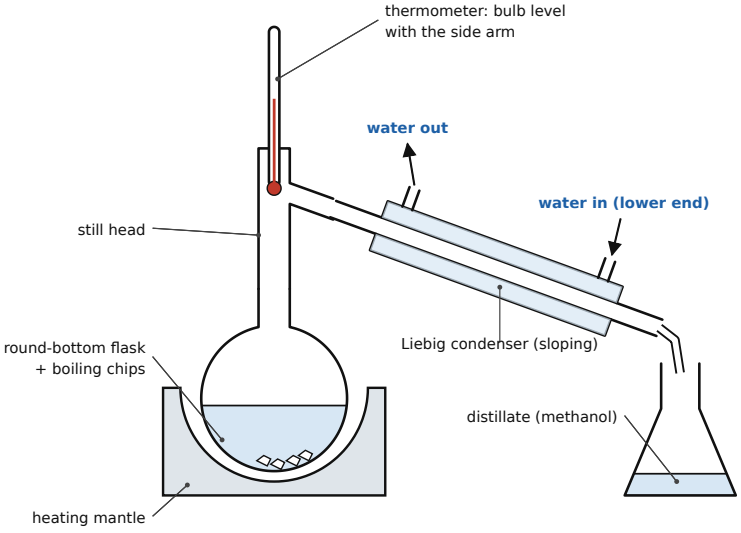
19 Heating under reflux

Used in: Organic WE6 (methyl cinnamate), WE7 (aspirin)

MUST SHOW

- Condenser fitted **vertically** on the round-bottom flask.
- Water **in at the bottom**, out at the top.
- Top left **open** (a sealed system could explode).
- Safe heating: heating mantle or water/oil bath, not a naked flame. Boiling chips.

Practise: cover the drawing and redraw it from the checklist.



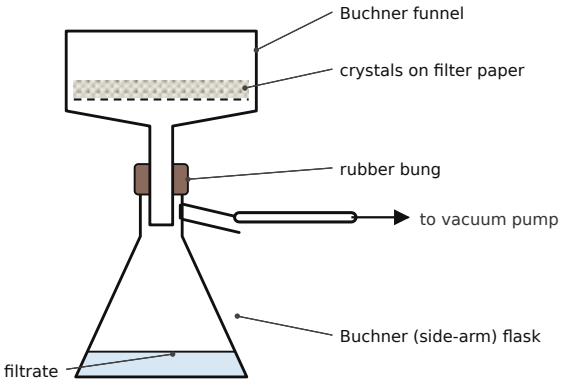
20 Distillation

Used in: Organic WE6 (recover methanol at 65 °C)

MUST SHOW

- Thermometer bulb **level with the side arm** (measures the vapour boiling off).
- Condenser sloping down to the receiver; water **in at the lower end**.
- Collect the fraction at its boiling point (methanol 65 °C).
- Heating mantle; boiling chips.

Practise: cover the drawing and redraw it from the checklist.



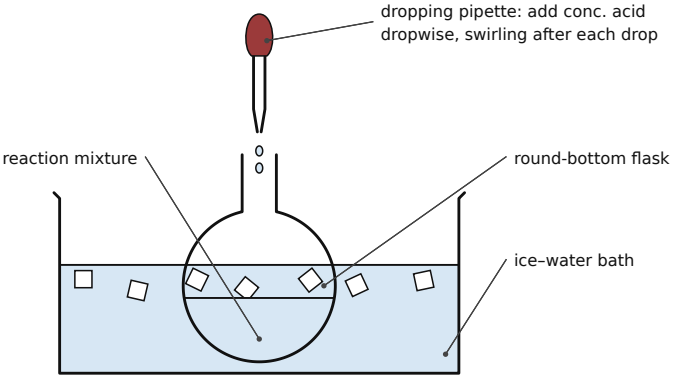
21 Vacuum filtration

Used in: Organic: four main steps · WE7 (aspirin)

MUST SHOW

- Buchner funnel with filter paper, sealed into a side-arm flask by a bung.
- Side arm connected to a vacuum pump.
- Wet the paper, apply the vacuum, pour quickly, wash the crystals with cold solvent.

Practise: cover the drawing and redraw it from the checklist.



22 Cooling while adding a reagent dropwise

Used in: Organic WE6 (conc. H_2SO_4), WE7 (H_3PO_4)

MUST SHOW

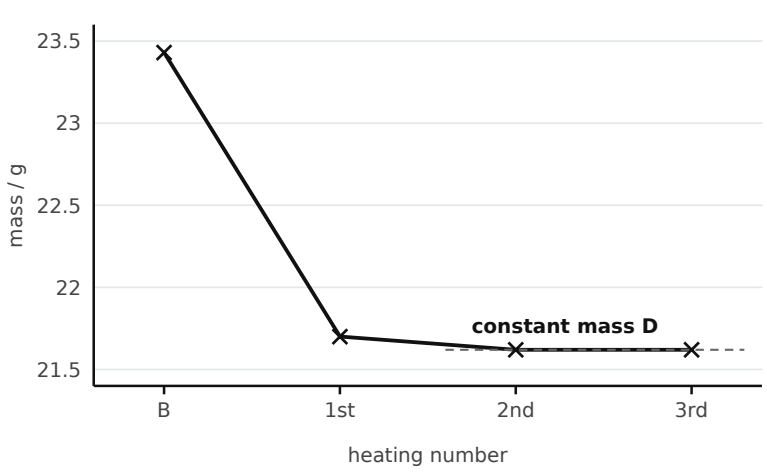
- Flask standing in an **ice–water bath**.
- Reagent added **dropwise** (dropper or dropping funnel) with swirling or stirring.
- Why: the addition is exothermic; cooling and slow addition limit the heat released and side reactions.

Practise: cover the drawing and redraw it from the checklist.

PART B

Graphs to sketch

GRAVIMETRY



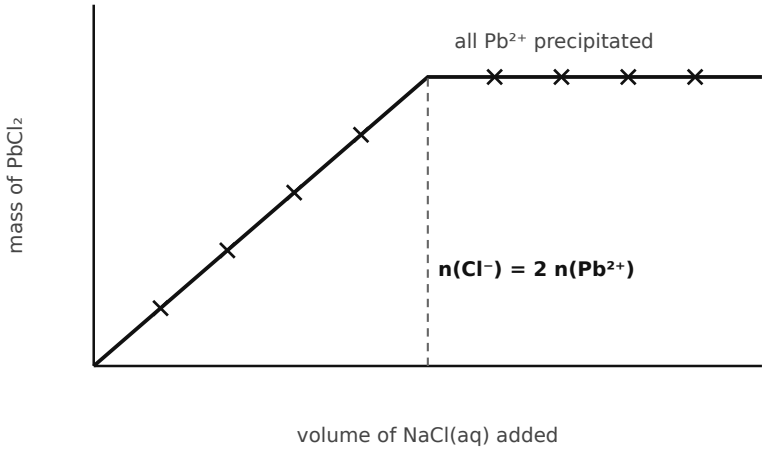
23 Mass against number of heatings

Used in: Gravimetry WE1, WE2, WE3

MUST SHOW

- Mass falls, then levels off: **constant mass** (D = D).
- Constant mass means decomposition (or drying) is complete.

Practise: cover the drawing and redraw it from the checklist.



24 Mass of precipitate against volume added

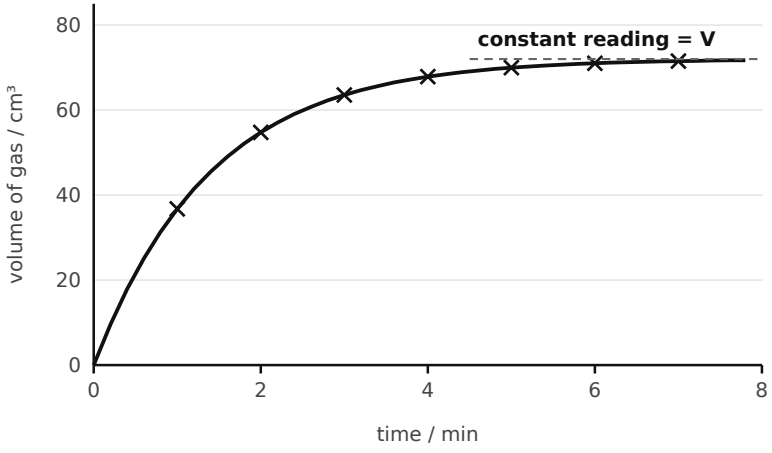
Used in: Introduction Ex 2 ($PbCl_2$ stoichiometry)

MUST SHOW

- Straight line up, then horizontal once all Pb^{2+} has precipitated.
- The break point gives the mole ratio.

Practise: cover the drawing and redraw it from the checklist.

GAS COLLECTION



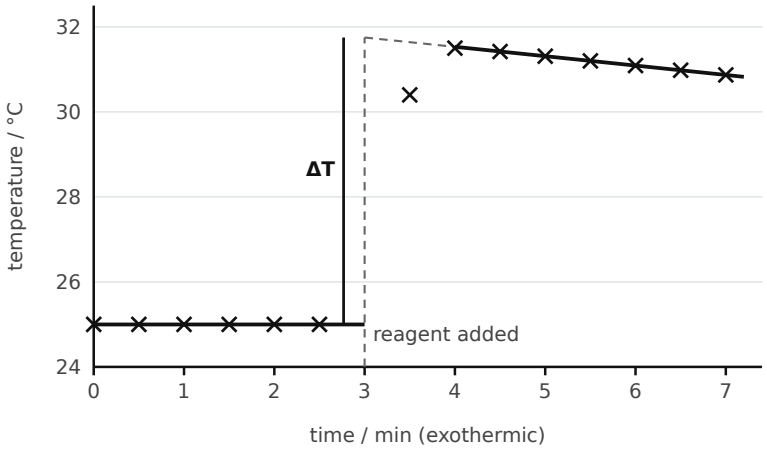
25 Gas volume against time

Used in: Gas collection: reading until constant

MUST SHOW

- Volume rises quickly, then levels off when the reaction stops.
- Record V only when the syringe reading is constant.

Practise: cover the drawing and redraw it from the checklist.



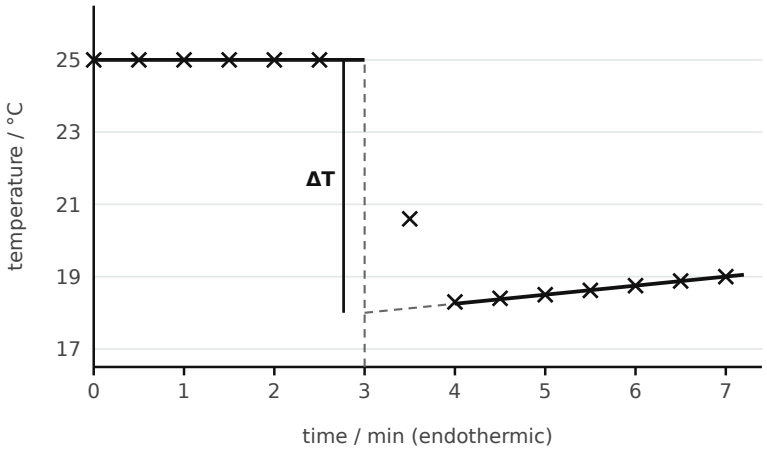
26 Temperature against time (exothermic)

Used in: Energetics: temperature–time method

MUST SHOW

- ≥ 3 readings before mixing at 3.0 min; ≥ 5 after the maximum.
- Two best-fit lines, both **extrapolated to t = 3.0 min**; ΔT is the vertical gap.

Practise: cover the drawing and redraw it from the checklist.



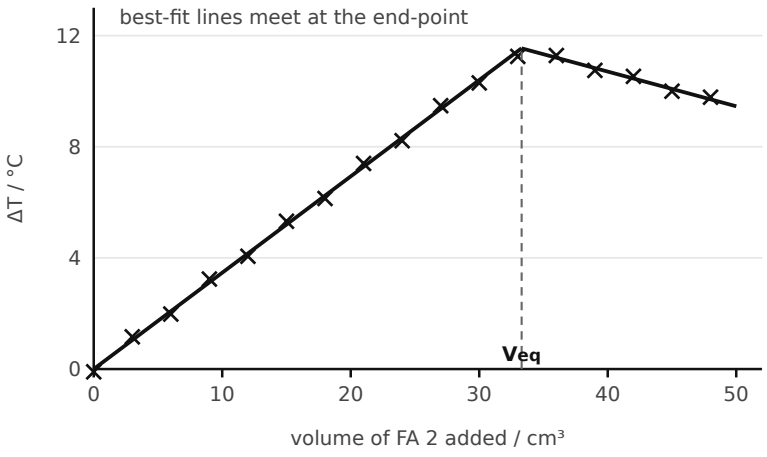
27 Temperature against time (endothermic)

Used in: Energetics: temperature–time method

MUST SHOW

- Same method; the solution warms back up after the minimum.
- ΔT is negative.

Practise: cover the drawing and redraw it from the checklist.



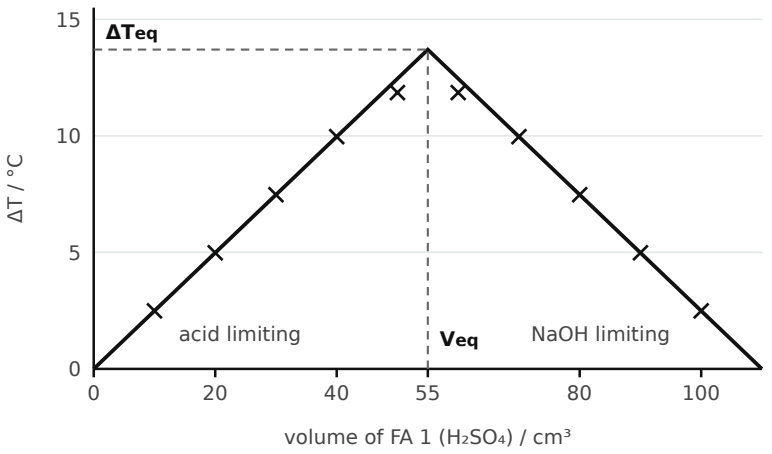
28 Thermometric titration

Used in: Energetics WE7

MUST SHOW

- Rising line, then falling line; they intersect at the end-point, V_{eq}.
- Label the end-point on the volume axis.

Practise: cover the drawing and redraw it from the checklist.



29 Mixing varying volumes (total constant)

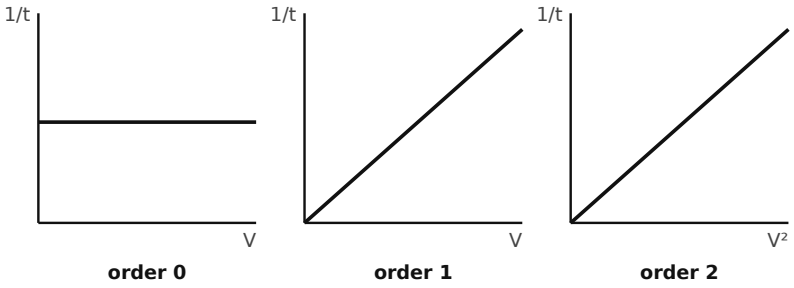
Used in: Energetics WE8

MUST SHOW

- Peak at the stoichiometric ratio: label V_{eq} and ΔT_{eq}.
- Left side: acid limiting; right side: alkali limiting.

Practise: cover the drawing and redraw it from the checklist.

KINETICS



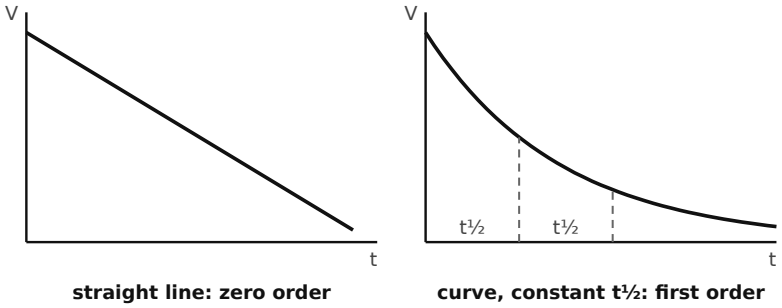
30 1/t against V: finding the order

Used in: Kinetics WE10 (clock method)

MUST SHOW

- Horizontal → order 0. Straight line through origin (1/t vs V) → order 1. Straight line through origin (1/t vs V^2) → order 2.

Practise: cover the drawing and redraw it from the checklist.



31 V($KMnO_4$) against time

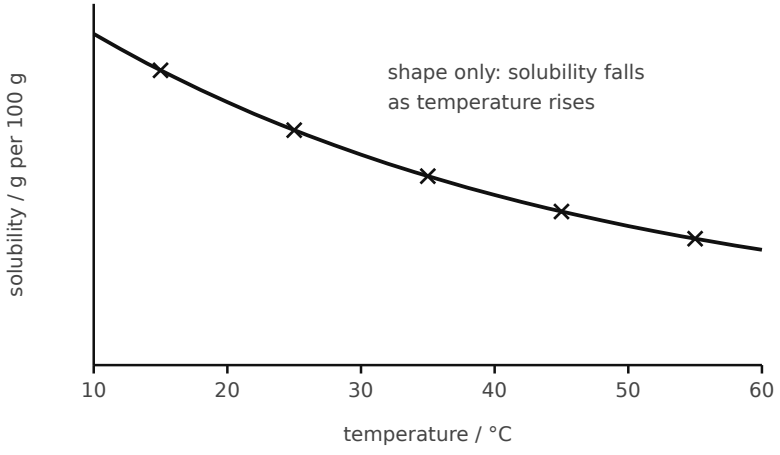
Used in: Kinetics WE11 (sampling method)

MUST SHOW

- Straight line down → zero order.
- Curve with constant half-life → first order.

Practise: cover the drawing and redraw it from the checklist.

EQUILIBRIA



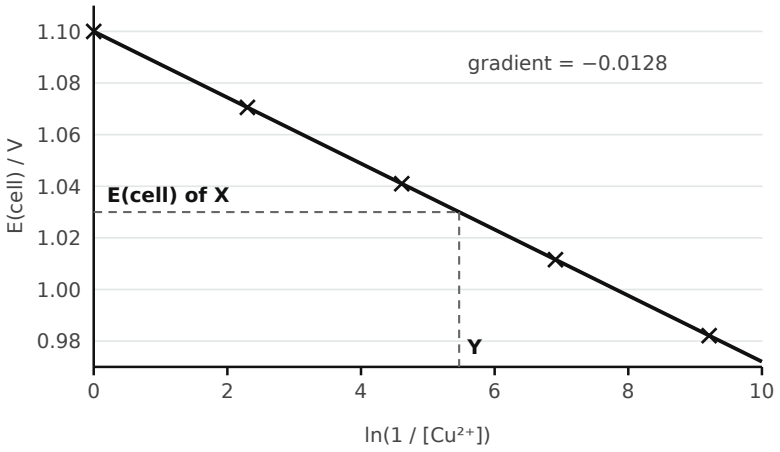
32 Solubility against temperature

Used in: Equilibria WE2 ($Ce(SO_4)_2$)

MUST SHOW

- Five temperatures, one point each; smooth curve.
- For $Ce(SO_4)_2$ the curve slopes down (solubility falls as T rises).

Practise: cover the drawing and redraw it from the checklist.



33 E(cell) against ln(1/[Cu^{2+}])

Used in: Electrochemistry WE5

MUST SHOW

- Straight line, gradient −0.0128 (from the Nernst equation).
- Read Y for the unknown X; $[Cu^{2+}] = 1/e^{Y}$.

Practise: cover the drawing and redraw it from the checklist.